

Bayesian Inference

Principles, Methods, and Applications

A Graduate-Level Introduction

Prepared for Graduate Students in Quantitative Fields

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Outline

- 1 Motivation & Philosophy
- 2 Bayes' Theorem
- 3 Prior Distributions
- 4 Conjugate Analysis
- 5 Posterior Inference
- 6 Hierarchical Models
- 7 Computational Methods
- 8 Model Comparison

Two Schools of Thought

Frequentist View

- Probability = *long-run frequency*
- Parameters θ are *fixed, unknown* constants
- Inference via confidence intervals & p -values
- Data are random; θ is not

Bayesian View

- Probability = *degree of belief*
- Parameters θ are *random variables*
- Inference via posterior distributions
- Prior knowledge is *formally encoded*

Key Insight

Bayesian inference provides a **coherent, probabilistic framework** for updating beliefs in the light of evidence.

Why Bayesian?

- ① **Principled uncertainty quantification** — posterior gives a full probability distribution over θ , not just a point estimate.
- ② **Sequential updating** — today's posterior becomes tomorrow's prior as new data arrive.
- ③ **Small-sample validity** — priors regularise estimates when data are scarce.
- ④ **Coherence** — decisions derived from posteriors satisfy the axioms of rational choice (de Finetti, Savage).
- ⑤ **Natural model comparison** — marginal likelihoods provide automatic Occam's razor.

Bayes' Theorem — Core Formula

Theorem (Bayes, 1763)

For parameter θ and data $\mathcal{D}$:

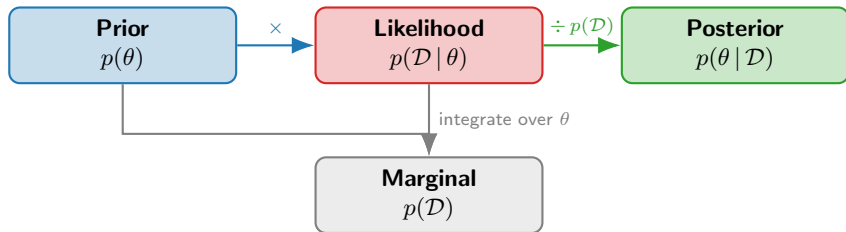
$$\underbrace{p(\theta | \mathcal{D})}_{\text{posterior}} = \frac{\overbrace{p(\mathcal{D} | \theta)}^{\text{likelihood}} \cdot \overbrace{p(\theta)}^{\text{prior}}}{\underbrace{p(\mathcal{D})}_{\text{marginal likelihood}}}$$

- $p(\theta)$ — encodes *prior* knowledge before seeing data.
- $p(\mathcal{D} | \theta)$ — the *likelihood* of observing $\mathcal{D}$ given θ .
- $p(\mathcal{D}) = \int p(\mathcal{D} | \theta) p(\theta) d\theta$ — normalising constant.
- $p(\theta | \mathcal{D})$ — updated belief after observing data.

Proportional form (sufficient for inference):

$$p(\theta | \mathcal{D}) \propto p(\mathcal{D} | \theta) p(\theta)$$

Visual Decomposition of Bayes' Theorem



The posterior is proportional to likelihood times prior; the marginal likelihood normalises.

Choosing a Prior

Types of priors:

- **Informative** — encodes genuine prior knowledge (e.g. from meta-analyses)
- **Weakly informative** — regularising; prevents pathological estimates
- **Non-informative / flat** — $p(\theta) \propto 1$; caution: can be improper
- **Conjugate** — chosen so that the posterior is in the same family as the prior (*analytical tractability*)
- **Jeffreys** — invariant under reparametrisation:
 $p(\theta) \propto \sqrt{\det \mathcal{I}(\theta)}$

Common Conjugate Pairs

Likelihood	Prior
Bernoulli / Binomial	Beta
Poisson	Gamma
Normal (σ^2 known)	Normal
Normal (μ known)	Inv-Gamma
Multinomial	Dirichlet

Beta Prior Family

Definition:

$$p(\theta; \alpha, \beta) = \frac{\theta^{\alpha-1}(1-\theta)^{\beta-1}}{B(\alpha, \beta)}, \quad \theta \in [0, 1]$$

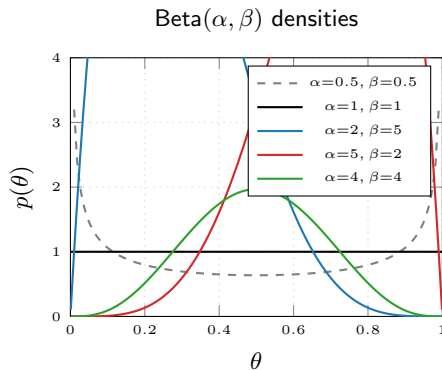
with $B(\alpha, \beta) = \Gamma(\alpha)\Gamma(\beta)/\Gamma(\alpha + \beta)$.

Moments:

$$\mathbb{E}[\theta] = \frac{\alpha}{\alpha + \beta}, \quad \text{Var}[\theta] = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}$$

Special cases:

- $\alpha = \beta = 1$: Uniform $[0, 1]$
- $\alpha = \beta$: symmetric about $\frac{1}{2}$



Beta-Binomial Conjugate Model

Setup: Estimate coin-flip probability θ .

Prior: $\theta \sim \text{Beta}(\alpha_0, \beta_0)$

Likelihood: $X \mid \theta \sim \text{Bin}(n, \theta)$

Posterior: $\theta \mid X=k \sim \text{Beta}(\alpha_0 + k, \beta_0 + n - k)$

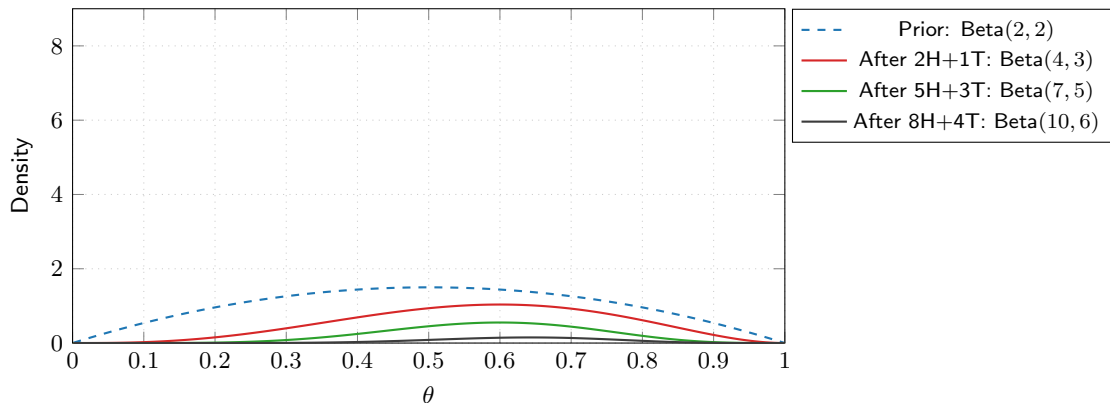
Derivation (proportional form)

$$p(\theta \mid k) \propto \theta^k (1 - \theta)^{n-k} \cdot \theta^{\alpha_0-1} (1 - \theta)^{\beta_0-1} = \theta^{(\alpha_0+k)-1} (1 - \theta)^{(\beta_0+n-k)-1}$$

- α_0, β_0 act as *pseudo-counts* from prior experience.
- **Posterior mean:** $\hat{\theta} = \frac{\alpha_0 + k}{\alpha_0 + \beta_0 + n}$ — a *shrinkage* estimator toward prior mean $\frac{\alpha_0}{\alpha_0 + \beta_0}$.
- As $n \rightarrow \infty$, posterior concentrates around MLE k/n .

Sequential Updating Illustrated

Sequential Beta-Binomial updating ($\alpha_0=2, \beta_0=2$)



Prior Beta(2, 2) updated after each batch of coin flips. Posterior sharpens and shifts toward the observed frequency.

Normal–Normal Conjugate Model

Setup: Known variance σ^2 ; unknown mean μ .

$$\text{Prior: } \mu \sim \mathcal{N}(\mu_0, \tau_0^2)$$

$$\text{Likelihood: } X_i \mid \mu \stackrel{iid}{\sim} \mathcal{N}(\mu, \sigma^2), \quad i = 1, \dots, n$$

$$\text{Posterior: } \mu \mid \mathbf{x} \sim \mathcal{N}(\mu_n, \tau_n^2)$$

where the posterior parameters are:

$$\frac{1}{\tau_n^2} = \frac{1}{\tau_0^2} + \frac{n}{\sigma^2}, \quad \mu_n = \tau_n^2 \left(\frac{\mu_0}{\tau_0^2} + \frac{n\bar{x}}{\sigma^2} \right)$$

Precision-weighted average interpretation

$$\mu_n = \underbrace{\frac{1/\tau_0^2}{1/\tau_0^2 + n/\sigma^2}}_{\text{prior weight}} \mu_0 + \underbrace{\frac{n/\sigma^2}{1/\tau_0^2 + n/\sigma^2}}_{\text{data weight}} \bar{x}$$

Posterior mean is a *precision-weighted* convex combination of prior mean and MLE.

Point Estimates from the Posterior

Given posterior $p(\theta | \mathcal{D})$, common point summaries:

Posterior Mean (MMSE)

$$\hat{\theta}_{\text{mean}} = \mathbb{E}[\theta | \mathcal{D}] = \int \theta p(\theta | \mathcal{D}) d\theta$$

Minimises expected squared error loss.

Posterior Median

$\hat{\theta}_{\text{med}}$ s.t. $\Pr(\theta \leq \hat{\theta}_{\text{med}} | \mathcal{D}) = 0.5$. Minimises expected absolute error.

MAP Estimate

$$\hat{\theta}_{\text{MAP}} = \arg \max_{\theta} p(\theta | \mathcal{D})$$

Equivalent to penalised MLE:

$$\hat{\theta}_{\text{MAP}} = \arg \max_{\theta} [\log p(\mathcal{D} | \theta) + \log p(\theta)]$$

ℓ_2 prior $\Rightarrow$ Ridge; ℓ_1 prior $\Rightarrow$ Lasso.

Credible Intervals

A $100(1 - \alpha)\%$ **credible interval** $[a, b]$ satisfies:

$$\Pr(a \leq \theta \leq b \mid \mathcal{D}) = 1 - \alpha$$

Two common choices:

- 1 **Equal-tailed interval (ETI):** $\alpha/2$ probability in each tail.
- 2 **Highest Posterior Density (HPD):** Shortest interval containing $1 - \alpha$ of the mass.

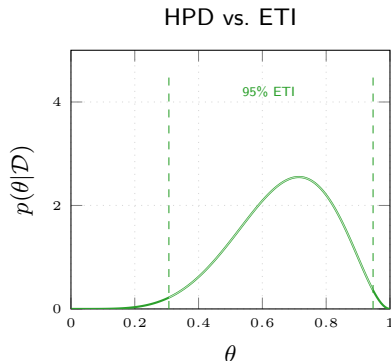
$$\{\theta : p(\theta \mid \mathcal{D}) \geq h_\alpha\}$$

HPD $\equiv$ ETI for symmetric unimodal posteriors.

Contrast with frequentist CI

A 95% CI does *not* mean “ θ is in this interval with prob. 95%”.

A 95% credible interval *does*.



Posterior Beta(6, 3); shaded = 95% ETI.

Posterior Predictive Distribution

Goal: Predict a new observation $\tilde{x}$ after seeing $\mathcal{D}$.

$$p(\tilde{x} | \mathcal{D}) = \int p(\tilde{x} | \theta) p(\theta | \mathcal{D}) d\theta$$

- Marginalises over all parameter uncertainty — *honest* uncertainty.
- Automatically *heavier tailed* than predictions at a fixed θ .
- Contrast: plug-in prediction uses $p(\tilde{x} | \hat{\theta})$ only.

Example — Beta-Binomial

Model: $X_i \sim \text{Bernoulli}(\theta)$, prior $\theta \sim \text{Beta}(\alpha, \beta)$.

Posterior predictive for $\tilde{X}$:

$$\Pr(\tilde{X} = 1 | \mathcal{D}) = \int_0^1 \theta \cdot \frac{\theta^{\alpha'-1} (1-\theta)^{\beta'-1}}{B(\alpha', \beta')} d\theta = \frac{\alpha'}{\alpha' + \beta'}$$

where $\alpha' = \alpha + \sum x_i$, $\beta' = \beta + n - \sum x_i$.

Hierarchical (Multi-Level) Models

Motivation: Data come from J related groups. Full pooling ignores group structure; no pooling ignores sharing. **Partial pooling** via hierarchy.

Two-level model:

$$y_{ij} \mid \theta_j \sim p(y \mid \theta_j)$$

$$\theta_j \mid \phi \sim p(\theta \mid \phi)$$

$$\phi \sim p(\phi)$$

Joint posterior:

$$p(\boldsymbol{\theta}, \phi \mid \mathbf{y}) \propto p(\phi) \prod_{j=1}^J p(\theta_j \mid \phi) \prod_i p(y_{ij} \mid \theta_j)$$

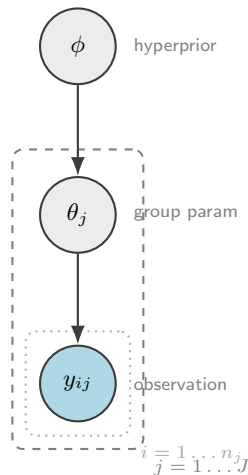


Plate diagram for a two-level hierarchical model.

Stein's Paradox & Shrinkage

- **James–Stein (1961):** For $J \geq 3$ groups, the pooled shrinkage estimator *dominates* the MLE in MSE.
- Hierarchical Bayes provides a rigorous probabilistic justification:

$$\hat{\theta}_j^{\text{EB}} = (1 - B) \bar{y}_j + B \hat{\mu}, \quad B = \frac{\hat{\sigma}_\epsilon^2 / n_j}{\hat{\sigma}_\theta^2 + \hat{\sigma}_\epsilon^2 / n_j}$$

- Groups with fewer observations are *shrunk more* toward the grand mean $\hat{\mu}$ — automatic regularisation.
- Full Bayes integrates over hyperparameters ϕ rather than plugging in estimates (*empirical Bayes*).

Practical Impact

Hierarchical models are ubiquitous in ecology, education research, clinical trials (multi-site), and sports analytics.

The Computational Challenge

- In closed-form only for conjugate models.
- For general models, the normalising constant $p(\mathcal{D}) = \int p(\mathcal{D} | \theta) p(\theta) d\theta$ is *intractable* (high-dimensional integral).
- Solution: **approximate** or **simulate** the posterior.

MCMC

Exact (asymptotic)

MH, HMC, Gibbs

Var. Inference

Fast approx.

ELBO, mean-field

Laplace Approx.

Normal approx.

mode + curvature

Markov Chain Monte Carlo (MCMC)

Idea: Construct a Markov chain with stationary distribution $p(\theta | \mathcal{D})$; run it long enough to obtain approximate samples.

Metropolis–Hastings:

- 1 Propose $\theta^* \sim q(\cdot | \theta^{(t)})$
- 2 Compute acceptance ratio:

$$r = \frac{p(\theta^* | \mathcal{D}) q(\theta^{(t)} | \theta^*)}{p(\theta^{(t)} | \mathcal{D}) q(\theta^* | \theta^{(t)})}$$

- 3 Accept $\theta^{(t+1)} = \theta^*$ with probability $\min(1, r)$;
else $\theta^{(t+1)} = \theta^{(t)}$

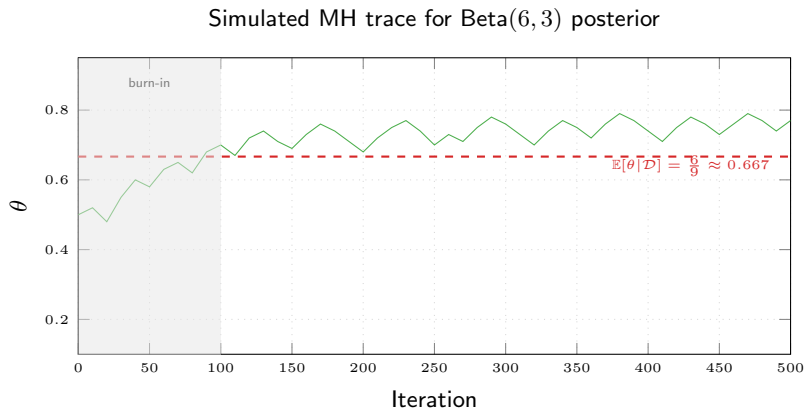
Hamiltonian Monte Carlo (HMC):

- Introduces auxiliary momentum $\mathbf{p}$
- Simulates Hamiltonian dynamics:

$$H(\theta, \mathbf{p}) = -\log p(\theta | \mathcal{D}) + \frac{1}{2} \mathbf{p}^\top M^{-1} \mathbf{p}$$

- Leapfrog integration preserves volume
- Dramatically reduces random-walk behaviour
- Basis of **Stan**, PyMC, NumPyro

MCMC Trace & Convergence



Discard *burn-in* samples; monitor $\hat{R}$ (Gelman–Rubin) across chains for convergence.

Variational Inference (VI)

Idea: Approximate intractable posterior with a tractable family $\mathcal{Q}$.

$$q^*(\theta) = \underset{q \in \mathcal{Q}}{\operatorname{argmax}} \operatorname{KL}(q(\theta) \parallel p(\theta \mid \mathcal{D}))$$

Equivalently, **maximise the ELBO** (Evidence Lower BOund):

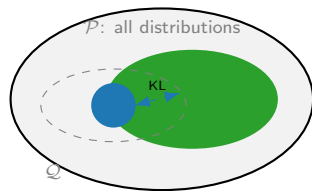
$$\mathcal{L}(q) = \mathbb{E}_{\theta \sim q} [\log p(\mathcal{D} \mid \theta)] - \operatorname{KL}(q(\theta) \parallel p(\theta)) \leq \log p(\mathcal{D})$$

Mean-field VI:

$$q(\theta) = \prod_{k=1}^K q_k(\theta_k)$$

- Coordinate-ascent updates (CAVI)
- Scalable to large datasets via SVI (stochastic)
- Underestimates posterior variance

KL divergence geometry:



Bayesian Model Comparison

Given models $\mathcal{M}_1, \mathcal{M}_2$, compare via the **Bayes factor**:

$$\text{BF}_{12} = \frac{p(\mathcal{D} \mid \mathcal{M}_1)}{p(\mathcal{D} \mid \mathcal{M}_2)} = \frac{\int p(\mathcal{D} \mid \theta, \mathcal{M}_1) p(\theta \mid \mathcal{M}_1) d\theta}{\int p(\mathcal{D} \mid \theta, \mathcal{M}_2) p(\theta \mid \mathcal{M}_2) d\theta}$$

Jeffreys' Scale of Evidence

$\log_{10} \text{BF}_{12}$	BF_{12}	Evidence for $\mathcal{M}_1$
$0 - \frac{1}{2}$	1–3.2	Barely worth mentioning
$\frac{1}{2} - 1$	3.2–10	Substantial
1–2	10–100	Strong
> 2	> 100	Decisive

- BF automatically penalises complex models (**Occam's razor**).
- Posterior odds = $\text{BF}_{12} \times$ prior odds.

Information Criteria

Practical approximations to cross-validation / marginal likelihood:

$$\text{DIC} = \overline{D(\theta)} + p_D, \quad p_D = \overline{D(\theta)} - D(\bar{\theta})$$

$$\text{WAIC} = -2 \sum_{i=1}^n \log \mathbb{E}[p(y_i | \theta)] + 2 \sum_{i=1}^n \text{Var}[\log p(y_i | \theta)]$$

$$\text{LOO-CV} \approx \sum_{i=1}^n \log p(y_i | \mathcal{D}_{-i})$$

- **DIC** (Spiegelhalter et al. 2002) — easy, but questionable for non-normal posteriors.
- **WAIC** (Watanabe 2010) — fully Bayesian; uses full posterior.
- **LOO-CV** — gold standard; approximated cheaply via Pareto-smoothed importance sampling (PSIS-LOO, Vehtari et al. 2017).

Worked Example: Clinical Trial

Problem: A new drug is given to $n = 40$ patients; $k = 28$ respond. Estimate the response rate θ and test $H_0 : \theta \leq 0.5$.

Model:

$$\theta \sim \text{Beta}(2, 2) \quad (\text{weakly informative})$$

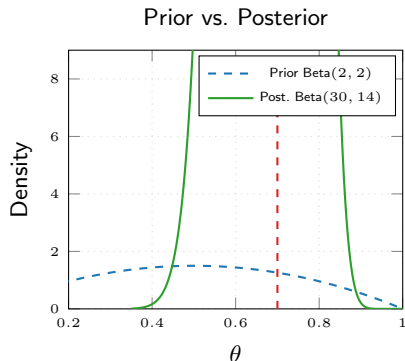
$$k \mid \theta \sim \text{Bin}(40, \theta)$$

Posterior:

$$\theta \mid k=28 \sim \text{Beta}(30, 14)$$

Summary:

- Posterior mean: $30/44 \approx 0.682$
- 95% ETI: $[0.53, 0.81]$
- $\Pr(\theta > 0.5 \mid k=28) \approx 0.997$



Bernstein–von Mises Theorem

Theorem (Bernstein–von Mises)

Under regularity conditions, as $n \rightarrow \infty$:

$$\sqrt{n}(\theta - \hat{\theta}_{\text{MLE}}) \mid \mathcal{D} \xrightarrow{d} \mathcal{N}(0, \mathcal{I}(\theta_0)^{-1})$$

where $\mathcal{I}(\theta_0)$ is the Fisher information at the true θ_0 .

- **Implication:** In large samples, the posterior is approximately Gaussian regardless of the prior (*prior washes out*).
- The $100(1 - \alpha)\%$ credible interval and confidence interval *coincide asymptotically*.
- **Caveat:** Breaks down for non-regular models, infinite-dimensional parameters, and misspecified models.

Practical Takeaway

Bayesian and frequentist methods agree for large n ; differences matter most in *small-sample* or *high-dimensional* settings.

Laplace Approximation

Idea: Approximate the log-posterior by a second-order Taylor expansion around the MAP $\hat{\theta}$:

$$\log p(\theta | \mathcal{D}) \approx \log p(\hat{\theta} | \mathcal{D}) - \frac{1}{2}(\theta - \hat{\theta})^\top \underbrace{\left[-\nabla_{\theta}^2 \log p(\theta | \mathcal{D}) \Big|_{\hat{\theta}} \right]}_{=: A} (\theta - \hat{\theta})$$
$$\Rightarrow p(\theta | \mathcal{D}) \approx \mathcal{N}(\hat{\theta}, A^{-1})$$

Uses:

- Fast posterior approximation (one optimisation + one Hessian)
- Analytic marginal likelihood approximation:
$$\log p(\mathcal{D}) \approx \log p(\mathcal{D} | \hat{\theta}) + \log p(\hat{\theta}) + \frac{d}{2} \log(2\pi) - \frac{1}{2} \log \det A$$
- Basis for BIC: $\text{BIC} \approx -2 \log \mathcal{L}(\hat{\theta}) + d \log n$

Summary of Key Concepts

Core framework:

$$\underbrace{p(\theta|\mathcal{D})}_{\text{posterior}} \propto \underbrace{p(\mathcal{D}|\theta)}_{\text{likelihood}} \cdot \underbrace{p(\theta)}_{\text{prior}}$$

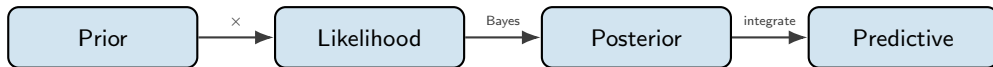
- Probability = rational degree of belief
- All inference flows from posterior
- Posterior predictive propagates uncertainty
- Sequential updating is natural

Computational toolkit:

- Conjugate priors — closed form
- Laplace — fast normal approx
- VI — scalable approximation
- MCMC (HMC) — gold standard

Model evaluation:

- Bayes factors
- WAIC / PSIS-LOO
- Posterior predictive checks



Further Reading

- [1] A. Gelman et al., *Bayesian Data Analysis*, 3rd ed., CRC Press, 2013.
- [2] C. Bishop, *Pattern Recognition and Machine Learning*, Springer, 2006.
- [3] R. McElreath, *Statistical Rethinking*, 2nd ed., CRC Press, 2020.
- [4] C. Robert, *The Bayesian Choice*, 2nd ed., Springer, 2007.
- [5] A. Vehtari, A. Gelman, J. Gabry, "Practical Bayesian model evaluation using leave-one-out cross-validation and WAIC," *Statistics and Computing*, 27(5), 2017.

Questions?

“Bayesian statistics is the analysis of uncertain knowledge by application of logic.”

— E.T. Jaynes

